Advanced high school · extension

Bézout and integer equations

Worked solutions

Exercise 1

Find gcd(84,30) and a Bézout identity.

$84 = 2 \times 30 + 24$ and $30 = 24 + 6$. Hence $6 = 3 \times 30 - 84$ and the gcd is 6.

Exercise 2

Does $12x + 18y = 5$ have integer solutions?

No: $gcd(12, 18) = 6$ does not divide 5.

Exercise 3

Find all integer solutions of $5x + 3y = 1$.

(−1,2) is a solution. All are $x = -1 + 3t, y = 2 - 5t$ for integer t.

Exercise 4

Among solutions of $5x + 3y = 31$, find those with nonnegative x and y.

Modulo $3, x \equiv 2$. Since $0 \leq x \leq 6, x$ is 2 or 5. The pairs are (2,7) and (5,2).

Problem: complete reasoning

An order of 100 items must be delivered in boxes of 7 and 11, without opening a box. Find every possible combination, then the one using the fewest boxes. Justify that no nonnegative solution has been missed.

Solve $7x + 11y = 100$. Modulo $7, 4y \equiv 2$, so $y \equiv 4$ (mod 7). Since $0 \leq y \leq 9$, only $y = 4$ works, giving $x = 8$. This requires 12 boxes. The solution is unique, so it automatically minimises the number of boxes.